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Hypothesis Testing



- THE **DECISION** ABOUT A **POPULATION** CAN BE TAKEN EASILY ON THE BASIS OF SAMPLES TAKEN FROM THE SAME POPULATION.
- TO MAKE DECISION ABOUT A POPULATION WE USE **SAMPLING THEORY**.

FOR EXAMPLE , If a drug chemist want to decide whether a new drug is effective or not in curing a disease. Then he can make judgement by using SAMPLING THEORY.

- SAMPLING THEORY INVOLVES THE **TESTS OF HYPOTHESIS**.

What is Hypothesis ?

- IN SAMPLING THEORY, TEST OF HYPOTHESIS ALWAYS BEGINS WITH AN ASSUMPTION.
- HYPOTHESES IS THE **ASSUMPTION** ABOUT THE **POPULATION PARAMETERS**.
- IN SAMPLING THEORY, IN ORDER TO MAKE A DECISION ABOUT A POPULATION FIRST OF ALL WE MAKE AN ASSUMPTION ABOUT THE POPULATION PARAMETERS.
- SUCH AN ASSUMPTION IS CALLED HYPOTHESIS.
- THIS HYPOTHESIS MAY BE TRUE OR MAY NOT BE TRUE.

What is Population Parameter ?

POPULATION PARAMETER IS THE VALUE WHICH DESCRIBES THE CHARACTERISTICS OF THE POPULATION.

FOR EXAMPLE :- POPULATION MEAN ' μ ', STANDARD DEVIATION OF POPULATION etc.

What is Sample Statistic ?

SAMPLE STATISTIC IS THE VALUE WHICH DESCRIBES THE CHARACTERISTICS OF THE SAMPLE .

FOR EXAMPLE :- SAMPLE MEAN, STANDARD DEVIATION OF SAMPLE etc.

TWO TYPES OF HYPOTHESIS:-

(1) NULL HYPOTHESIS

(2) ALTERNATIVE HYPOTHESIS

What is Null Hypothesis ?

- NULL HYPOTHESIS IS THE ASSUMPTION THAT THERE IS NO SIGNIFICANT DIFFERENCE BETWEEN THE POPULATION PARAMETERS AND THE SAMPLE STATISTICS.
- IN NULL HYPOTHESIS WE ASSUME THAT OUR ASSUMPTION IS TRUE AND TEST IS DONE FOR THE POSSIBILITY OF REJECTION OF NULL HYPOTHESIS.
- NULL HYPOTHESIS IS DENOTED BY H_0

The average age of university students is 22 years:

null hypothesis

$$H_0 = \mu = 22$$

- The **alternative hypothesis**, symbolized by H_1 , is a statistical hypothesis that states the existence of a **difference** between a parameter and a specific value, or states that there is a **difference** between two parameters.

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alternative hypothesis

$$H_1 = \mu \neq 22$$

To state the hypothesis, researchers must translate the conjecture or claim words into mathematical symbols. The following symbols are used in hypothesis testing. ✚

equal to $=$

less than $<$

greater than $>$

not equal to $\neq$

two-tailed test

$$H_0 = \mu = a$$

$$H_1 = \mu \neq a$$

left-tailed test

$$H_0 = \mu = a$$

$$H_1 = \mu < a$$

right-tailed test

$$H_0 = \mu = a$$

$$H_1 = \mu > a$$

NULL HYPOTHESIS AND ALTERNATIVE HYPOTHESIS ARE OPPOSITE TO EACH OTHER . IT MEANS,

IF NULL HYPOTHESIS IS TRUE THEN ALTERNATIVE HYPOTHESIS IS FALSE.

IF NULL HYPOTHESIS IS FALSE THEN ALTERNATIVE HYPOTHESIS IS TRUE.

Part 2

Concept of errors in test of hypothesis

**Type 1 errors
and
Type 2 errors**

- **IN THE PROCESS OF TEST OF HYPOTHESIS SOME ERRORS MAY OCCUR. THESE ERRORS ARE CALLED ERRORS IN HYPOTHESIS TESTING.**

- **TWO TYPES OF ERRORS IN HYPOTHESIS TESTING :-**

- 1. TYPE 1 ERRORS**

- 2. TYPE 2 ERRORS**

What is TYPE 1 Error ?

TYPE 1 ERROR OCCURS WHEN WE REJECT THE NULL HYPOTHESIS THOUGH IT IS TRUE .

WHEN NULL HYPOTHESIS IS TRUE AND WE MAKE THE DECISION OF REJECTING IT THEN IT IS TYPE 1 ERROR.

The Probability of TYPE 1 error is denoted by α

Thus, The Probability of not occurring TYPE 1 Error is $P = 1 - \alpha$

What is TYPE 2 Error ?

TYPE 2 ERROR OCCURS WHEN WE ACCEPT THE NULL HYPOTHESIS THOUGH IT IS FALSE .

WHEN NULL HYPOTHESIS IS FALSE AND WE MAKE THE DECISION OF ACCEPTING IT THEN IT IS TYPE 2 ERROR.

The Probability of TYPE 2 error is denoted by β

Thus, The Probability of not occurring TYPE 2 Error is $P = 1 - \beta$

	ACCEPT NULL HYPOTHESIS H_0	REJECT NULL HYPOTHESIS H_0
H_0 Is True	Correct Decision $P = 1 - \alpha$	Type 1 Error $P = \alpha$
H_0 Is False	Type 2 Error $P = \beta$	Correct Decision $P = 1 - \beta$

PART - 3

CONCEPT OF LEVEL OF SIGNIFICANCE, CONFIDENCE LEVEL AND CRITICAL REGION

What is level of Significance ?

THE LEVEL OF SIGNIFICANCE IS DENOTED BY α (alpha). α (alpha) REPRESENTS THE MAXIMUM PROBABILITY OF OCCURRING TYPE 1 ERROR.

LEVEL OF SIGNIFICANCE IS THE PROBABILITY OF REJECTING A NULL HYPOTHESIS WHEN IT IS TRUE .

GENERALLY IT IS TAKEN AT 5% LEVEL. IT SHOWS 95% PROBABILITY OF DECISION TO BE CORRECT.

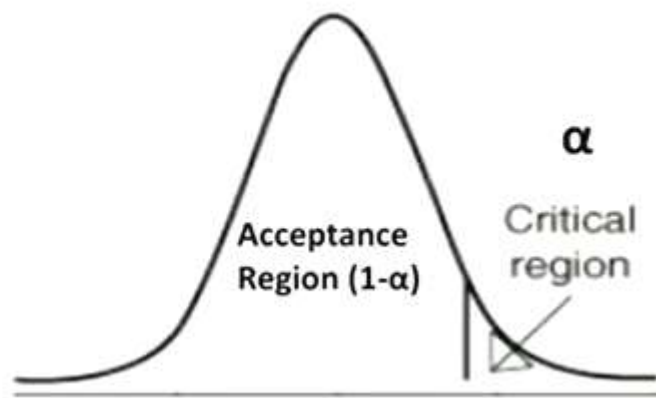
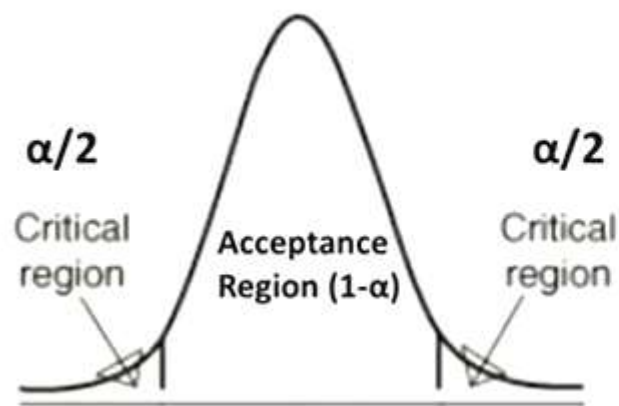
LEVEL OF SIGNIFICANCE MAY ALSO BE TAKEN AT 1% LEVEL. IT SHOWS 99% PROBABILITY OF DECISION TO BE CORRECT.

Confidence Level

- Confidence Level is the probability of Not Occuring Type 1 Errors.
- It is denoted by $1 - \alpha$
- Confidence Level is meant by Accepting Null Hypothesis when it is True.

What is Critical Region ?

CRITICAL REGION IS ALSO KNOWN AS REJECTION REGION. THE REGION (Area) UNDER THE NORMAL CURVE WHICH IS NOT COVERED BY REJECTION REGION IS CALLED ACCEPTANCE REGION.



PART - 4

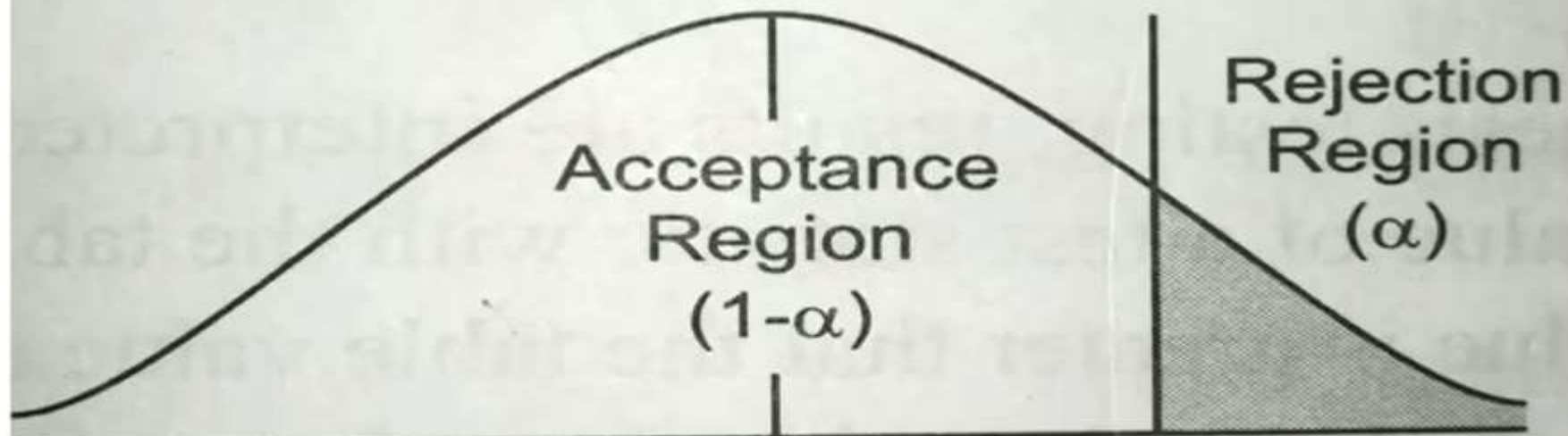
CONCEPT OF ONE TAILED TEST,
TWO TAILED TEST AND
CRITICAL VALUES IN TEST OF
HYPOTHESIS

What is One Tailed Test ?

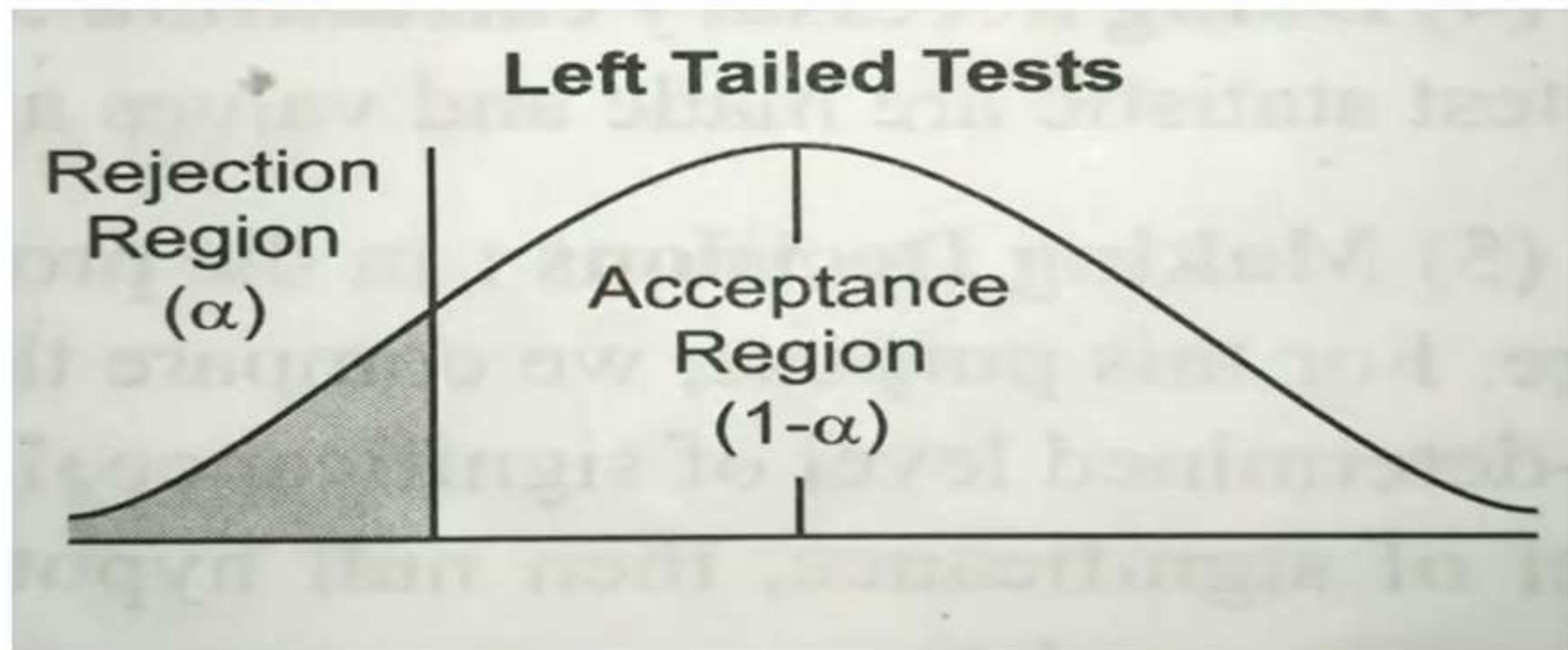
- ONE TAILED TEST IS A TEST OF HYPOTHESIS IN WHICH ALTERNATIVE HYPOTHESIS IS EXPRESSED BY USING SYMBOL OF LESS THAN ($<$) OR MORE THAN ($>$).
- IN THIS TEST OF HYPOTHESIS ENTIRE CRITICAL REGION LIES IN THE ONE TAIL OF DISTRIBUTION CURVE.
- CRITICAL REGION WILL LIE AT THE RIGHT TAIL IF ALTERNATIVE HYPOTHESIS HAS SYMBOL OF MORE THAN ($>$)
- CRITICAL REGION WILL LIE AT THE LEFT TAIL IF ALTERNATIVE HYPOTHESIS HAS SYMBOL OF LESS THAN ($<$).

ACCEPTANCE REGION AND REJECTION REGION IN ONE TAILED TEST

Right Tailed Tests



ACCEPTANCE REGION AND REJECTION REGION IN ONE TAILED TEST

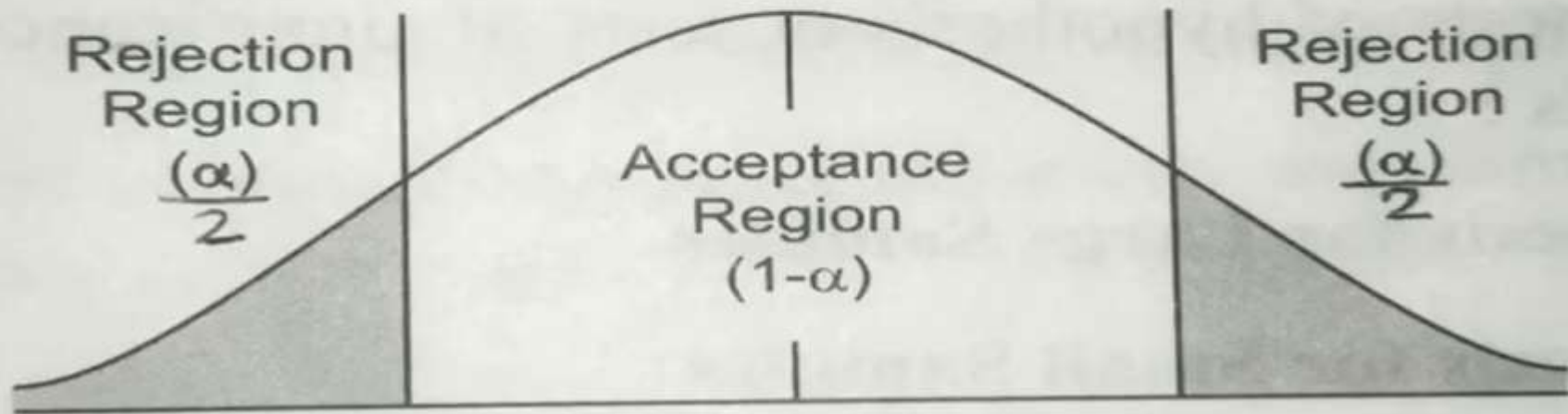


What is Two Tailed Test ?

- TWO TAILED TEST IS A TEST OF HYPOTHESIS IN WHICH ALTERNATIVE HYPOTHESIS IS EXPRESSED BY USING SYMBOL NOT EQUAL TO ($\neq$)
- IN THIS TEST OF HYPOTHESIS CRITICAL REGION IS DEVIDED INTO TWO PARTS. THUS CRITICAL REGION LIES AT BOTH TAILS (LEFT TAIL & RIGHT TAIL).

ACCEPTANCE REGION AND REJECTION REGION IN TWO TAILED TEST

Two Tailed Tests
(l.o.s. α)



Case 1

The average age of university students is 22 years:



Note: This is called a **two-tailed** hypothesis test.

The hypothesis for this situation are

null hypothesis

alternative hypothesis

$$H_0 = \mu = 22$$

$$H_1 = \mu \neq 22$$

What is Critical Value ?

- In hypothesis testing, a critical value is a point on the test distribution that is compared to the test statistic to determine whether to reject the null hypothesis.
- If the absolute value of test statistic is greater than the critical value, we will reject the null hypothesis.
- If the absolute value of test statistic is less than the critical value, we will accept the null hypothesis.

Part – 5

Procedure for Test of Hypothesis

Step 1 Set up a Null Hypothesis

Step 2 Set up a suitable level of Significance

Step 3 Set up a Suitable Test of Hypothesis

Step 4 Calculation for Test Statistics

Step 5 Compare Calculated Value of Test Statistics with Table Value

Step 6 Take Suitable Decisions

STEP 1

SET UP A NULL HYPOTHESIS

- **NULL HYPOTHESIS IS DENOTED BY H_0**

IT ASSUMES THAT THERE IS NO SIGNIFICANT DIFFERENCE BETWEEN THE POPULATION

PARAMETERS (Population Mean, Population Standard Deviation etc.) **AND THE SAMPLE STATISTICS** (Sample Mean, Sample Standard Deviation etc.).

STEP 2

SET UP A SUITABLE LEVEL OF SIGNIFICANCE

Level of Significance is denoted by α (alpha). α (alpha) represents the maximum probability of Occuring Type 1 Errors.

If No level of Significance is given in the question then 5% Significance level is used.

STEP 3

SET UP A SUITABLE TEST OF HYPOTHESIS

Test of Hypothesis can be done by using any of the Test like:- Z Test, t Test, F Test etc.

The Selection of Suitable Test depends upon the available information.

for example :-

- when number of observation in sample is more than 30 ($n > 30$) then we will use Z test
 - but if number of observations in samples are less than 30 ($n < 30$) and population parameters are not known then we will use student's t test
-

Step 4

Calculation for Test Statistics

After selecting appropriate Test, computations for calculating the test statistic are made and values are calculated.

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

z test for the mean when σ is given

Step 5

Compare calculated values of test statistics with the table values

In this step test statistics are compared with the critical values at pre determined level of Significance.

Step 6

Taking Suitable Decisions

- If the calculated value of test statistics is greater than the critical values then reject the null hypothesis.
- If the calculated value of test statistics is less than the critical values then accept the null hypothesis.

Finding the Critical Values for Specific α Values, Using Table:



Draw the figure and indicate the appropriate area.

1. If the test is left-tailed, the critical region, with an area equal to α , will be on the left side of the mean.
2. If the test is right-tailed, the critical region, with an area equal to α , will be on the right side of the mean.
3. If the test is two-tailed, α must be divided by 2: one-half of the area will be to the right of the mean, and one-half will be to the left of the mean.

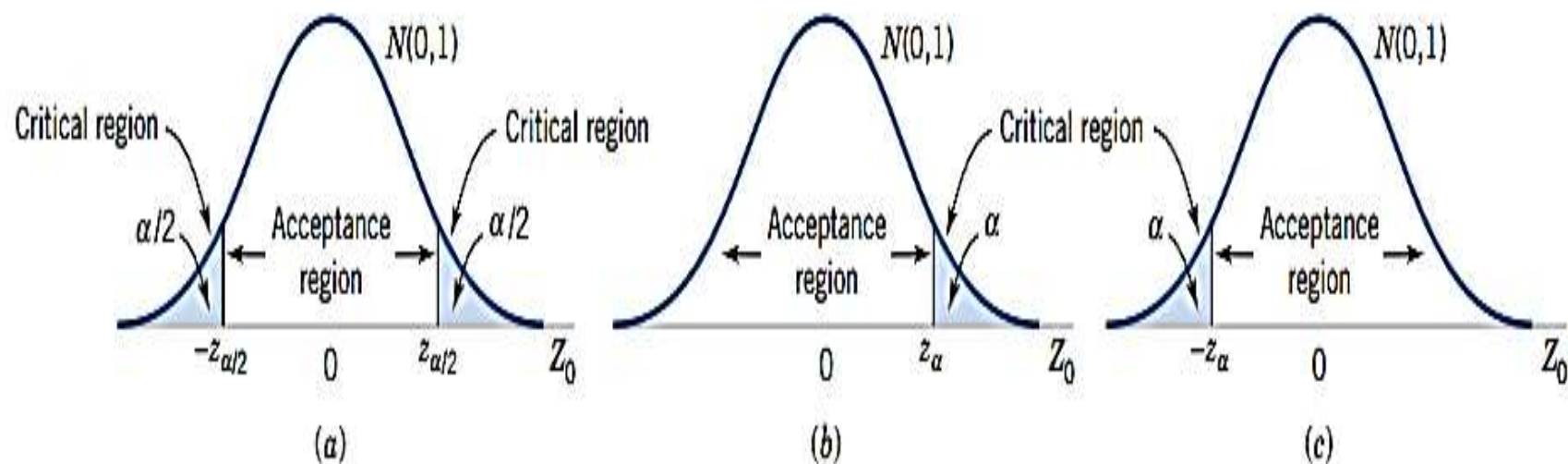


Figure 9-6 The distribution of Z_0 when $H_0: \mu = \mu_0$ is true, with critical region for (a) the two-sided alternative $H_1: \mu \neq \mu_0$, (b) the one-sided alternative $H_1: \mu > \mu_0$, and (c) the one-sided alternative $H_1: \mu < \mu_0$.

1. For a left-tailed test, use the z value that corresponds to the area equivalent to α in Table.



2. For a right-tailed test, use the z value that corresponds to the area equivalent to $1 - \alpha$.

3. For a two-tailed test, use the z value that corresponds to $\alpha/2$ for the left value. It will be negative. For the right value, use the z value that corresponds to the area equivalent to $1 - \alpha/2$. It will be positive.

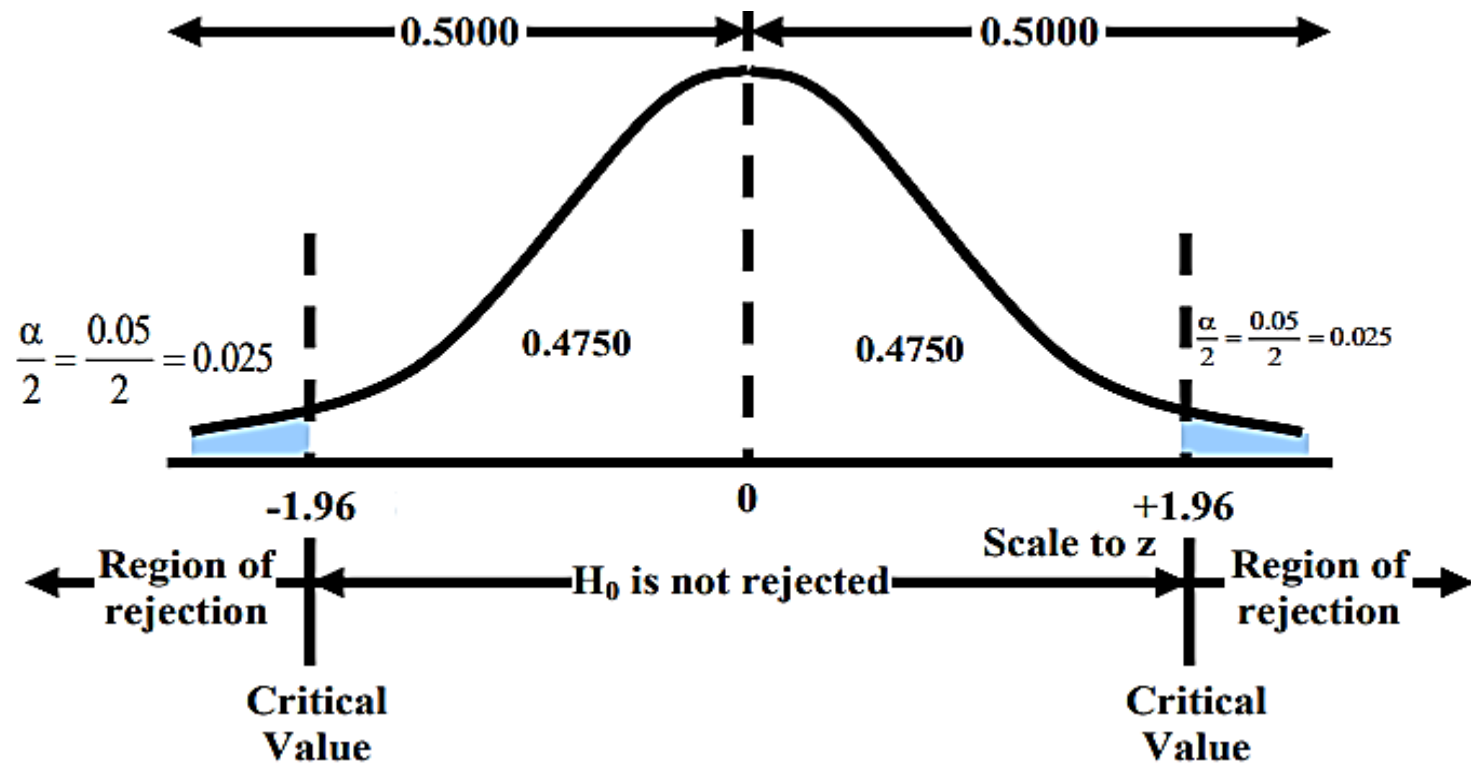
Question

Aircrew escape systems are powered by a solid propellant. The burning rate of this propellant is an important product characteristic. Specifications require that the mean burning rate must be 50 centimeters per second. We know that the standard deviation of burning rate is 2 centimeters per second. The experimenter decides to specify a type I error probability or **significance level of $\alpha = 0.05$** and selects a random sample of $n = 25$ and obtains a **sample average burning rate is 51.3** of centimeters per second. What conclusions can be drawn ?

1. The parameter of interest is μ , the mean burning rate.
2. $H_0: \mu = 50$ centimeters per second
3. $H_1: \mu \neq 50$ centimeters per second
4. $\alpha = 0.05$

$\bar{x} = 51.3$ we specified $\alpha = 0.05$ and so the boundaries of the critical region are at $\frac{\alpha}{2} = 0.025$

$$z_{0.025} = 1.96 \quad \text{and} \quad -z_{0.025} = -1.96$$



for right tail = $1 - \frac{\alpha}{2} = 0.025$ using z -table we get 1.96

$$z_0 = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

$$z_0 = \frac{51.3-50}{\frac{2}{\sqrt{25}}} = 3.25$$

Conclusion: Since $z_0 = 3.25 > 1.96$, we reject $H_0: \mu = 50$ at the 0.05 level of

9-29. Medical researchers have developed a new artificial heart constructed primarily of titanium and plastic. The heart will last and operate almost indefinitely once it is implanted in the patient's body, but the battery pack needs to be recharged about every four hours. A random sample of 50 battery packs is selected and subjected to a life test. The average life of these batteries is 4.05 hours. Assume that battery life is normally distributed with standard deviation 0.2 hour.

(a) Is there evidence to support the claim that mean battery life exceeds 4 hours? Use $\alpha = 0.05$.

a) 1) The parameter of interest is the true average battery life, μ .

2) $H_0 : \mu = 4$

3) $H_1 : \mu > 4$

4) $\alpha = 0.05$

5) $z_0 = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

6) Reject H_0 if $z_0 > z_{\alpha}$ where $z_{0.05} = 1.65$

7) $\bar{x} = 4.05$, $\sigma = 0.2$

6) Reject H_0 if $z_0 > z_{\alpha}$ where $z_{0.05} = 1.65$

7) $\bar{x} = 4.05$, $\sigma = 0.2$

$$z_0 = \frac{4.05 - 4}{0.2 / \sqrt{50}} = 1.77$$

8) Since $1.77 > 1.65$, reject the null hypothesis and conclude that there is sufficient evidence to conclude that the true average battery life exceeds 4 hours at $\alpha = 0.05$.

Practice Problem 1:

Using the z Table find the critical value for each:

- | | |
|---|----------------------|
| 1. $\alpha = 0.05$, left tailed test | -1.65 Answers |
| 2. $\alpha = 0.05$, two tailed test | +,- 1.96 |
| 3. $\alpha = 0.01$, left tailed test | -2.33 |
| 4. $\alpha = 0.005$, right tailed test | +2.58 |
| 5. $\alpha = 0.01$, right tailed test | +2.33 |
| 6. $\alpha = 0.02$, left tailed test | -2.05 |
| 7. $\alpha = 0.05$, right tailed test | +1.65 |

P-value method for hypothesis testing

When we perform a hypothesis test in statistics, a p -value helps us determine the significance of the results.

The p -value is used as an alternative to rejection points to provide the smallest level of significance at which the null hypothesis would be rejected.

Hypothesis Testing

Decision rule when using a P-value

- If **P-value $\leq \alpha$** , reject the null hypothesis
- If **P-value $> \alpha$** , do not reject the null hypothesis

Step 1 State the hypotheses and identify the claim.

Step 2 Compute the test value.

Step 3 Find the *P*-value.

Step 4 Make the decision.

Step 5 Summarize the results.

Smallest level of significance = *p - value*

1. $1[1 - P(Z < a)]$ for two tail test. $H_0: \mu = a$, $H_1: \mu \neq a$
2. $1 - P(Z < a)$ for right tail test: $H_0: \mu = a$, $H_1: \mu = \mu_0 > a$
3. $P(Z < a)$ for left tail test $H_0: \mu = a$, $H_1: \mu = \mu_0 < a$

9-25. An engineer who is studying the tensile strength of a steel alloy intended for use in golf club shafts knows that tensile strength is approximately normally distributed with $\sigma = 60$ psi. A random sample of 12 specimens has a mean tensile strength of $\bar{x} = 3250$ psi.

- (a) Test the hypothesis that mean strength is 3500 psi. Use $\alpha = 0.01$.
- (b) What is the smallest level of significance at which you would be willing to reject the null hypothesis?
- (c) Explain how you could answer the question in part (a) with a two-sided confidence interval on mean tensile strength.

a) 1) The parameter of interest is the true mean tensile strength, μ .

2) $H_0 : \mu = 3500$

3) $H_1 : \mu \neq 3500$

4) $\alpha = 0.01$

5) $z_0 = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

6) Reject H_0 if $z_0 < -z_{\alpha/2}$ where $-z_{0.005} = -2.58$ or $z_0 > z_{\alpha/2}$ where $z_{0.005} = 2.58$

7) $\bar{x} = 3250$, $\sigma = 60$

$$z_0 = \frac{3250 - 3500}{60 / \sqrt{12}} = -14.43$$

8) Since $-14.43 < -2.58$, reject the null hypothesis and conclude the true mean compressive strength is significantly different from 3500 at $\alpha = 0.01$.

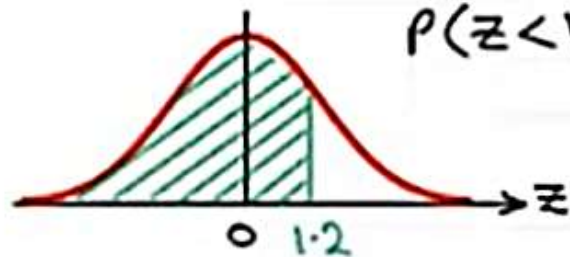
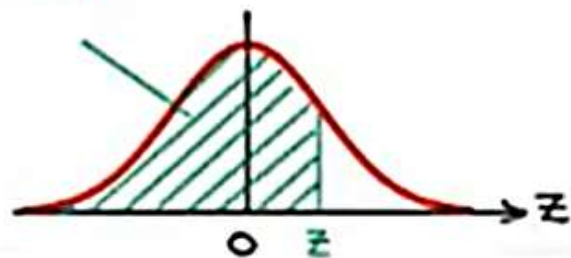
$$\begin{aligned}\text{Smallest level of significance} &= p\text{-value} = 1[1 - P(Z < 14.43)] \\ &= 2[1 - 1] = 0\end{aligned}$$

The smallest level of significance at which we are willing to reject the null hypothesis is 0.

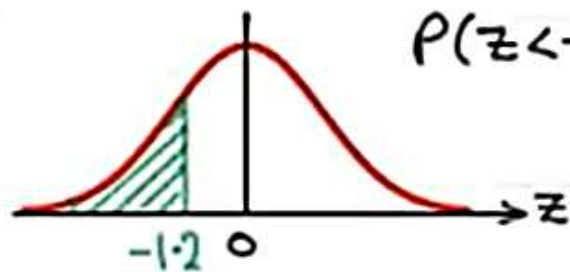
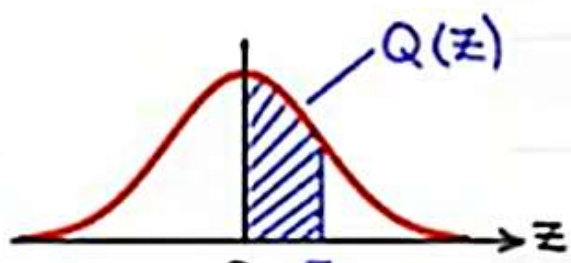
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$P(Z)$

$Z \sim N(0,1)$



$$P(Z < 1.2) = P(1.2) \\ = 0.88493$$



$$P(Z < -1.2) = P(-1.2) \\ = 0.11507$$

